

Divyam Sengar

San Diego, CA

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SKILLS

Java, Python, C, C++, Go, R, SQL, MATLAB, HTML/CSS, JavaScript, CUDA, JetBrains, Git, JUnit, PlayCrypt, React, Express, MongoDB, Node, Android, Roboelectric, RoomDatabase, Docker, GCP, Kubernetes, PyTorch, sklearn, TensorFlow, Pandas, Valgrind, GDB, Nachos, Requests, gRPC, Convex optimization, Transformer, TabTransformer, AWS, Numpy, Scipy

WORK EXPERIENCE

Computer Science and Engineering Society @ UCSD

Jun 2023 - Sep 2023

Full Stack Developer Intern

San Diego, CA

- Developed a dynamic membership platform website using React.js and Node.js, enabling secure user authentication and increasing membership engagement by 50%
- Implemented the backend with Express.js and MongoDB to efficiently store and manage user & event data

UC San Diego Computer Science and Engineering

Sep 2023 - Present

Instructional Assistant

San Diego, CA

- Mentored 500+ student classes for Python Programming, Advanced Data Structures, Computer Organization, and Deep Learning
- Conducted tutoring over 20+ weekly office hours, clarifying concepts and bolstering student coding proficiency
- Produced homework, exams, and dynamic question generators, improving assessment of student comprehension

DataAnnotation

Jun 2024 - Sep 2024

Software Engineer Intern

Remote

- Optimized AI chatbot performance by designing and implementing fine-tuning strategies for transformer-based models, resulting in improved code correctness and efficiency.
- Engineered and validated training pipelines for large language models, focusing on the generation of diverse and challenging coding problems across multiple languages to enhance model reasoning.
- Conducted in-depth analysis of model architecture, identifying key bottlenecks and implementing data-driven optimization techniques to reduce latency and improve output quality.
- Evaluated the performance of various chatbot models with respect to differential coding tasks, focusing on key performance metrics such as code correctness, brevity, and execution time.

Amazon Web Services

Jun 2025 - Sep 2025

Software Development Engineer Intern

Bellevue, WA

- Developing and integrating an automation chatbot for the AWS OpenSearch Release Optimization Team
- Customized Chatbot NLP Architecture, configured MCP Servers for real time data streaming, and implemented custom OpenSearch VectorDB based RAG Architecture for accurate LLM knowledge retrieval
- Wrote Infrastructure as Code (IaC) for AWS components (dynamoDB, bedrock foundational models, agents, api gateways) for one click deployment and CI/CD updates
- Configured agent-to-agent (a2a) protocol to communicate between workflow automation agents and chatbot, automating release tasks

PROJECTS

Data Compressor | C++

- Developed a custom Huffman file compression & decompression tool in C++ for efficient, lossless data compression
- Minimized compression overhead and optimized compression speed by utilizing custom data structures
- Reduced file size by 60% on average, applying a frequency based encoding approach

QuoteGuard | Python, PyTorch

- Developed a custom transformer-classifier to verify speakers of political quotes & speeches
- Achieved 96.2% accuracy by leveraging custom sparse attention mechanisms and positional embeddings
- Optimized accuracy via regularization, cross-validation, and varying the feed forward layer normalization functions

Diabetes Prediction | Python, sklearn, Pandas

- Codeveloped a diabetes prediction model using physiological and demographic data, achieving 96.3% accuracy
- Implemented and optimized SVM, XGBoost, Logistic Regression, and Random Forest Classifier models
- Optimized model performance via hyperparameter tuning and cross-validation techniques to reduce overfitting
- Decimated false positives via data augmentation to engineer cross-validation datasets

Successorator | Java, SQL, Android, Git, Github Actions, XML

- Designed and implemented a multifaceted to-do list Android app using Java, with recurrent tasks, focus mode, categories, calendar synchronization, persistence, deletion, rollover, and task preemption
- Used SQL & RoomDatabase to implement persistent features, speedy data retrieval, and enable task rollover logic
- Customized MVP architecture to streamline app workflows, ensuring intuitive and fast user input
- Built a robust Github Actions test suite, ensuring proper merge protection and automated testing

Welp | Golang, Python, Docker, Kubernetes, GCP, gRPC

- Built a Yelp-like microservices platform with scalable Detail, Review, and Reservation services using Golang and gRPC, deployed via Docker/Kubernetes with automated CI/CD pipelines
- Designed latency-throughput analysis, identifying frontend bottlenecks and improving max throughput by 4.2x via Kubernetes pod scaling.
- Integrated LRU/LFU caching policies to reduce storage-layer access latency by 85% under Zipfian workloads, achieving sub-10ms cache-hit response times
- Engineered a C-based thread-scheduling benchmark to optimize cache locality, reducing cross-core memory latency by 10.8ms (GCP) and cache misses via same-core co-location

Privacy Impact Indicator & AI Platform Privacy Analysis

- Designed and developed the Privacy Impact Indicator, a Chrome MV3 extension that visualizes real-time web tracking risk using an exponential privacy scoring model
- Engineered a system integrating service workers, ring-buffered telemetry, and peripheral UI feedback, nudging users away from tracker-heavy sites with no additional workload
- Conducted a 15-participant within-subjects study, reducing tracker contact by 32% and third-party requests by 14%, with high user-rated usefulness and minimal perceived intrusiveness
- Built a Selenium-Scrapy web crawler to audit cookie practices on major AI platforms (like ChatGPT & Deepseek), analyzing SameSite, Secure, and HttpOnly attributes over 157 cookies
- Discovered widespread use of long-lived first-party cookies (avg. 293 days) and incomplete security flag adoption, highlighting a critical need for modernized web privacy controls

Thread Affinity Analyzer | C, Linux, Bash, Perf

- Developed a C-based benchmarking utility to analyze the performance of multithreaded concurrency primitives under different CPU scheduling scenarios.
- Engineered a test suite to quantify the impact of thread-to-core affinity on shared memory access, measuring and diagnosing performance bottlenecks through cache-miss rates and CPU cycle analysis.
- Demonstrated a deep understanding of OS fundamentals by programmatically manipulating thread affinity with Linux system calls (pthread_setaffinity_np) to control scheduling behavior.
- Validated that co-locating threads on the same core significantly improves cache locality and reduces memory access latency: a core principle in high-performance system designs.

EDUCATION

University of California, San Diego
Bachelor of Science in Computer Science

Sep 2022 - Mar 2025

University of California, San Diego
Master of Science in Computer Science

Jan 2025 - Mar 2026

Total Relevant Coursework:

Advanced Data Structures, Algorithm Design & Analysis, Computability & Complexity, Computer Architecture, Computer Networks, Computer Security, Convex Optimization Formulations and Algorithms, Data Center Systems, Deep Learning, Digital Systems, Discrete & Continuous Optimization, Human-AI Interaction, Modern Cryptography, Numerical Analysis: Approximation & Nonlinear Equations, Numerical Linear Algebra, Object Oriented Development, Operating Systems, Parallel Computing, Programming Languages, Quantum Complexity & Cryptography, Quantum Computing, Recommender Systems & Web Mining, Software Engineering, Statistical NLP, Theory of Computation, Machine Learning with Music, Probabilistic AI, CyberSecurity & Privacy